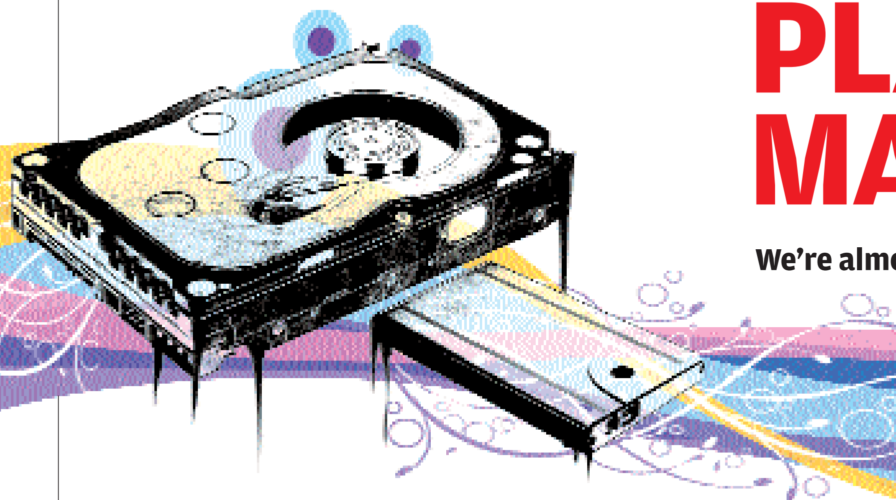

PLATTER MATTERS

We're almost there—in terabyte territory!

Over the past year we've seen an almost complete shift to Serial ATA. Manufacturers have also been bumping up both drive speeds and storage size. Several new models offer the best of both these worlds. Even better, you needn't pay much—hard drives are seriously affordable today!

Sceptics—would you call a 320 GB hard drive retailing at Rs 5,000 cheap, or what? 250 GB hard drives hover around the 4,000 rupee mark! The reason for this price downswing is the sharp incline in demand; multimedia content, DVD movies, and games being the reasons.

Myths And Realities

- **Choose any hard disk, it won't affect performance:** Wrong, big time! A hard drive is the slowest performing device in your PC, and its biggest bottleneck. The speed of a hard drive's spindle together with the integrated drive electronics affects performance a great deal.
- **Bigger hard disks are much costlier, overlook them:** Not so; it's price per GB you should look at. A 250 GB hard disk may have a lower cost per GB than a 160 GB drive.
- **SATA II is much better than SATA I:** SATA II is a newer standard. Performance gains aren't that really much. The difference in actual, real-world data transfer rates would be less than ten per cent! The pricing should be the deciding factor.
- **Warranties make the big difference:** Today, warranties are identical across hard drive vendors for most models: five years. However, you do need to enquire about warranties, since some models—typically older ones—may come with shorter warranties. In any case, remember that if a crash happens, money isn't much of a consolation when you've lost your data!

What To Look For

SATA II
SATA II is allows a theoretical transfer rate of 300 MB per second, as opposed to SATA I which allows half as much. Although this isn't true in real-world scenarios, we've seen manufacturers up the storage solutions on their motherboards to utilise some of the extra bandwidth this new standard provides. SATA II drives also have support for Native Command Queuing, which improves drive performance during multiple access operations—when you multitask applications or do multiple copying and pasting.

Size
Define your usage. Since 160 GB drives are dirt-cheap, it doesn't make sense buying anything smaller. If your DVD and music collection is extensive, you may need something in the range of 250 GB. Working with space-hungry RAW video files may see the need for a 500 GB drive. Simple browsing and office work may never need more than 80 GB.

Parallel ATA
Don't waste your time here. Parallel ATA hard disks are only for you if your motherboard doesn't support Serial ATA.

Specifications
Spindle speed really makes a difference—the faster the better. 7200 rpm is *de facto* now, although you do get 10,000 rpm drives aimed at gamers and enthusiasts. Buffer doesn't make as big a difference, and 8 MB of buffer should be sufficient. Stay away from any 5400 rpm drives!

Multiple drives
Multiple drives will perform better if the load is shared. If your OS resides on one drive, and program files and data reside on the other, you'll notice improved performance. However, you'll also need more room in your cabinet and more power to drive all of them. The heat generated by three working hard drives will also be more.

The newer, bigger drives today are much faster than their predecessors of even a year ago. So a 250 GB drive today will be better than two older, comparable 80 GB drives. However, two 250 GB drives will perform better than a single drive provided the swap file and applications are on one drive and the OS on the other.

Future Trends

Besides the performance advantages of Serial ATA technology, we've also seen the introduction of Perpendicular Recording. This technology allows the packing of much more data (increased areal density) onto the same platter size.

We've also witnessed the dawn of a rather intriguing technology—HAMR. Heat Assisted Magnetic Recording uses lasers in conjunction with different variants of the media being used today to dramatically increase the amount of data that a drive can hold. Promises of hard drives capable of storing 10 or more terabytes of data five or six years down the line is something worth looking at!

Although the hard disk is still by far the slowest performer in your PC, it's reassuring to see the firm steps being taken to remedy the situation. ■